MATH 5023

Project 8: Multivariable Taylor Theorem

Main source. C. H. Edwards, Jr., *Advanced Calculus of Several Variables*, Chapter II.

- Let $U \subset \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}$.
- $f \in C^k(U)$ means all iterated partial derivatives of order at most k exist and are continuous on U.
- $D_i f = \dfrac{\partial f}{\partial x_i}$. Thus $D_{i_1} D_{i_2} \cdots D_{i_q} f$ means repeated partial differentiation.
- For a fixed direction $h = (h_1, \ldots, h_n)$, the directional derivative operator is

$$D_h = h_1 D_1 + \cdots + h_n D_n.$$

- $D_h^r f(a)$ means apply the same directional derivative r times and then evaluate at a.

- $f \in C^2(U) \Rightarrow f_x, f_y, f_{xx}, f_{xy}, f_{yx}, f_{yy}$ exists.
- iterated partial derivative : $f(x,y) = x^2y + 3y^2$.

$D_2 f = f_y = x^2 + 6y$, $D_1 D_2 f = \frac{\partial}{\partial x}(x^2 + 6y) = 2x$

- $f(x,y) = x^2 + y^2$, $h(3,4)$, $D_1 f = 2x$, $D_2 f = 2y$, $D_h = D(3,4) = 3D_1 + 4D_2$

$= 6x + 8y$

- $D_h^2 f = D_h(6x + 3y) = 3 \cdot D_1(6x + 8y) + 4 \cdot D_2(6x + 8y)$

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Multivariable Taylor Theorem

Let $k \geq 0$, $a \in \mathbb{R}^n$, $h \in \mathbb{R}^n$, and

$$L = \{a + th : 0 \leq t \leq 1\}.$$

Suppose $U \subset \mathbb{R}^n$ is open, $L \subset U$, and $f : U \to \mathbb{R}$ has continuous partial derivatives up to order $k+1$. Then there exists $\xi \in L$ such that

$$f(a+h) = \sum_{r=0}^{k} \frac{D_h^r f(a)}{r!} + \frac{D_h^{k+1} f(\xi)}{(k+1)!}.$$

$P_k(h)$ $\qquad$ $R_k(h)$

$f(x,y) = x^2 + xy + y^3$, $a = (0,0)$, $h = (u,v) = a+h$.

Since $k=2$, $r = 0, 1, 2$.

$$P_2(u,v) = \frac{f(a)}{0!} + \frac{D_h f(a)}{1!} + \frac{D_h^2 f(a)}{2!}$$

$\vdots$ skip formula

$$= u^2 + uv$$

$$f(a+h) = f(u,v) = u^2 + uv + v^3$$

$$f(a+h) = P_2(h) + R_2(h)$$

$P_2(h) = u^2 + uv$ $\qquad$ $R_2(h) = v^3$

Step 1: Reduce to One Variable

↙ Similar to proof of M.V.T.

Define

$$g : [0,1] \to \mathbb{R}, \qquad g(t) = f(a + th).$$

Then

$$g(0) = f(a), \qquad g(1) = f(a + h).$$

By Chain rule, $g'(t) = D_1 f(a+th) h_1 + \cdots + D_n f(a+th) h_n$

$= D_h f(a+th)$

Repeating same process r times, we get

$g^{(r)}(t) = D_h^r f(a+th)$

where $g^{(r)}(t) = g(t)$, for $r = 0, 1, \cdots, k+2$.

Step 2: Substitute into Taylor Theorem

Apply the one-variable Taylor theorem:

$$g(1) = \sum_{r=0}^{k} \frac{g^{(r)}(0)}{r!} + \frac{g^{(k+1)}(c)}{(k+1)!}$$

for some $c \in (0,1)$.

Notice, $g(1) = \sum_{r=0}^{k} \frac{g^{(r)}(0)}{r!} + \frac{g^{(k+1)}(c)}{(k+1)!}(b-a)^{k+1}$

$g^{(r)}(0) = D_h^r f(a)$ $\quad g^{(k+1)}(c) = D_h^{k+1} f(a+ch)$

set $\xi = a+ch$, then $\xi \in L$. $\Rightarrow$ $f(a+h) = \sum_{r=0}^{k} \frac{D_h^r f(a)}{r!} + \frac{D_h^{k+1} f(\xi)}{(k+1)!}$

$P_k(h)$

$\Rightarrow$ remainder term is

$R_k(h) = f(a+h) - P_k(h)$

$= \frac{D_h^{k+2} f(\xi)}{(k+1)!}$

Example

Let

$$f(x,y) = e^{x+y}, \qquad a = (0,0), \qquad h = (x,y).$$

Observe $k=2$, by T.F. $P_2(h) = f(a) + D_h f(a) + \frac{D_h^2 f(a)}{2!}$

$f(a)$: 1 ; $D_h f$: $x+y$

$D_h^2 f = D_h(D_h f) = \left(u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y}\right)\cdot(uf_x + vf_y)$

$= u\frac{\partial}{\partial x}(uf_x + vf_y) + v\frac{\partial}{\partial y}(uf_x + vf_y)$

$= u^2 f_{xx} + uvf_{xy} + uvf_{yx} + v^2 f_{yy}$

Since $f \in C^2$, $f_{xy} = f_{yx}$, plug in $(0,0)$. We get $D_h^2 f = (x+y)^2$

$\Rightarrow P_2(h) = 1 + (x+y) + \frac{(x+y)^2}{2}$. By T.F., $\exists\, \xi = (\xi_1, \xi_2)$ on line segment from $(0,0)$ to (x,y). Similar calculation, we can find remainder, this

$$e^{x+y} = 1 + (x+y) + \frac{(x+y)^2}{2} + \frac{e^{\xi_1+\xi_2}\cdot(x+y)^3}{6}$$

For instance, if $|x| \le 0.1$, $|y| \le 0.1$, then $|\xi_1 + \xi_2| \le 0.2$

$$\Rightarrow |R_2(h)| \le \frac{e^{0.2}\,|x+y|^3}{6}$$ ← bound.

Connection with Newton's Method

Newton's method uses the second-order Taylor approximation near the current point a:

$$f(a+h) \approx f(a) + \nabla f(a) \cdot h + \frac{1}{2} h^T H(a) h.$$

The idea is to choose h so that this quadratic approximation is as small as possible. Differentiating the quadratic approximation with respect to h gives

$$\nabla f(a) + H(a)h = 0.$$

If $H(a)$ is invertible, then

$$h = -H(a)^{-1} \nabla f(a), \qquad a_{\text{new}} = a + h.$$

Thus Newton's method is built from Taylor's theorem: it replaces f by its local quadratic model and moves to the minimum of that model.

- By T. theorem: $f(a+h) \approx f(a) + D_h f(a) + \frac{1}{2} D_h^2 f(a)$. Notice $D_h f(a) = \nabla f(a) \cdot h$

$D_h^2 f(a) = h^T H(a) h$

where $\nabla f(a)$ is gradient.

ex: $f(x,y)$. $\nabla f(a) = \begin{pmatrix} f_x(a) \\ f_y(a) \end{pmatrix}$

$H(a)$ is Hessian matrix,

$H(a) = \begin{pmatrix} f_{xx}(a) & f_{xy}(a) \\ f_{yx}(a) & f_{yy}(a) \end{pmatrix}$ it shows how much function curve i.e. curvature.

h^T is transpose, if $h = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$, then $h^T = (h_1 h_2)$